

Lecture 1

Introduction

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Why quantum computing?

Why we should care?

In simple terms, because a lot of really difficult, complex problems will forever remain **out of reach** of classical supercomputers ...

- Classical computer technology is running up against **fundamental size limitations**
(Recall Moore's law: the observation that the number of transistors in a dense integrated circuit doubles about every two years.)



- Quantum computing brings new ideas to handle some really difficult, complex problems, which remain **out of reach** of classical supercomputers.

Why quantum computing?

Prime factorization



- As factoring a 130 digit number takes around one month in a massively parallel computer network, a 400 digit number will take about the age of the universe (10^{10} years).
- However, a quantum algorithm exists such that factoring is achieved in polynomial time.

Moving fast ...

Quantum is moving (very) fast...

Research on quantum technologies is **speeding up**, and has already **created the first operational and commercially available applications**.

The quantum decade: the 2020s ...

- For the first time the viability of quantum computing is **demonstrated in a number of problems** and **its utility discussed across industries**.
- Efforts, at national or international levels, to further **scale up** this research and development are in place.
- A **cross-industry race has begun** to secure quantum talent, build quantum skills, map real-world problems to quantum algorithms, and capture quantum application intellectual property.

... but still uncertain what the future will bring

- Quantum computing will have a **substantial impact on societies**:
- ... indeed, questions on its impact have changed from **if** to **when** and **how**,
- ... even if its **commercial potential** in the near term (5 to 10 yrs) is still debatable.

What we know for sure

- Resorting to a so **radically different technology**, it is difficult to **anticipate its evolution**.
- Emerging as an **entirely new paradigm**, quantum hardware and software **brings no similarity to their classical counterparts**.

... but what are we talking about?

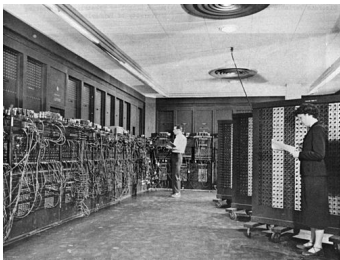
Where shall I begin, please Your Majesty?, he asked.

*Begin **at the beginning**, the King said gravely, **and go on till you come to the end: then stop.***

Lewis Carroll, *Alice's Adventures in Wonderland*, 1865

... the beginnig?

- **Information** has a crucial **physical** dimension
- **computation** is always a **physical** process



ENIAC (1946)



IBMQ (2022)

Quantum Mechanics ‘meets’ Computer Science

Two main intellectual achievements of the 20th century met

- Computer Science and Information theory progressed by **abstracting** from the physical reality. This was the key of its success to an extent that **its origin was almost forgotten**.
- On the other hand, **quantum mechanics** ubiquitously underlies ICT devices at the implementation level, but had no influence on the **computational model** itself ...
- ... until **now!**

Quantum Mechanics 'meets' Computer Science

Alan Turing (1912 - 1934)



On Computable Numbers, with an Application to the Entscheidungsproblem (1936)
(computability and the birth of computer science)

Quantum Mechanics 'meets' Computer Science

Richard Feynman (1918 - 1988)



Simulating Physics with Computers (1982)
(quantum reality as a computational resource)

Quantum effects as computational resources

Superposition

Our perception is that an object — e.g. a **bit** — exists in a well-defined state, even when we are not looking at it.

However: A quantum state **holds information of both possible classical states**, collapsing to one of them upon **observation**.

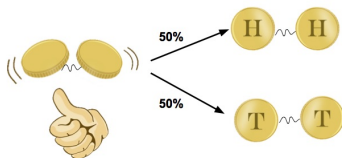


Quantum effects as computational resources

Entanglement

Our perception is that objects are directly affected only by nearby objects, i.e. the laws of physics work in a local way.

However: two qubits can be so strongly correlated, or **entangled**, that an action performed on one of them **can have an immediate effect on the other** even at a distance.



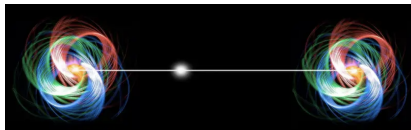
Quantum effects as computational resources

An **entangled** state cannot be considered as a result of the states of its individual constituents:

This phenomenon can be found in very simple mathematical structures: for example, binary relation

$$\{(0, 0), (1, 1)\}$$

cannot be written as a Cartesian product of subsets of $\mathcal{B} = \{0, 1\}$.



Quantum entanglement can be used to create instantaneous agreement on information across very long distances.

Quantum effects as computational resources

God plays dice indeed

Our perception is that the laws of Physics are deterministic: there is a unique outcome to every experiment.

However: Unlike classical physics, quantum theory has processes that are irreducibly **non-deterministic**, i.e. that cannot be accounted for solely by a lack of knowledge about reality. One can only know the **probability** of the outcome.

Uncertainty is a feature, not a bug

Our perception is that with better tools we will be able to measure whatever seems relevant for a problem.

However: there are **inherent limitations** to the amount of knowledge that one can ascertain about a physical system

Do you remember this cartoon?

Quantum is trendy ... but weird ...



Quantum Computation

David Deutsch (1953)



The Church-Turing principle and the universal quantum computer (1985)
(first example of a quantum algorithm that is exponentially faster than any possible deterministic classical one)

Quantum Computation

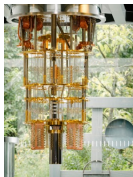
quantum resources



quantum algorithms



computability



Can we play the quantum game with a classical computer?

Do to superposition, a single quantum state (qubit) has a **continuum of possible values**, potentially storing lots of classical data.

Simulating a computation with qubits in a classical computer would be extremely hard, i.e. extremely inefficient as the number of qubits increases:

- For 100 qubits the state space would require to store $2^{100} \approx 10^{30}$ complex numbers!
- And what about rotating a vector in a vector space of dimension 10^{30} ?

Thus,

Quantum computing as **using quantum reality as a computational resource**

Richard Feynman, *Simulating Physics with Computers* (1982)

What can be expected from quantum computation?

- The meaning of **computable** remains the same ...
- ... but the order of **complexity** may change

The landmark

Factoring in **polynomial** time - $\mathcal{O}((\ln n)^3)$

Peter Shor, *Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer* (1994)

Which problems a Quantum Computer can solve?

- 1994: Peter Shor's factorization algorithm (exponential speed-up),
- 1996: Grover's unstructured search (quadratic speed-up),
- 2018: Advances in hash collision search, i.e finding two items identical in a long list — serious threat to the basic building blocks of secure electronic commerce.
- 2019: Google announced to have achieved quantum supremacy

Availability of proof of concept hardware

Explosion of emerging applications in several domains: security, finance, optimization, machine learning, ...

Where exactly do we stand?

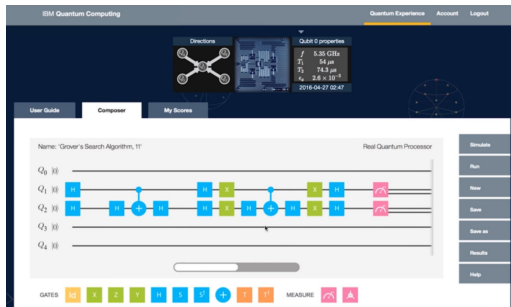
- Quantum devices have associated **decoherence times**, which limit the number of quantum operations that can be performed before the results are 'drowned' by noise.
- Each operation performed with quantum gates introduces **accuracy errors** in the system, which **limits the size of quantum circuits** that can be executed reliably.



Where exactly do we stand?

NISQ - Noisy Intermediate-Scale Quantum Hybrid machines:

- the quantum device as a coprocessor
- typically accessed as a service over the cloud



Still a long way to go ...

Historically, much of fundamental physics has been concerned with discovering the fundamental particles of nature and the equations which describe their motions and interactions.

It now appears that a different programme may be equally important: to discover the ways that nature allows and prevents information to be expressed and manipulated, rather than particles to move.

Steane, A.M., 1998.

Our programme

1. **Motivation: Quantum effects as computational resources.**
2. **Part 1: Quantum information and computation.**
 - 2.1 From bits to qubits: Single-state quantum systems.
 - 2.2 Superposition and interference.
 - 2.3 Hilbert spaces and the quantum state postulates: representation, composition, evolution, and measurement.
 - 2.4 Entanglement.
3. **Part 2: Quantum Algorithms.**
 - 3.1 Quantum gates and the circuit model
 - 3.2 Query algorithms
 - 3.3 Classical computations over the quantum circuit model
 - 3.4 Algorithms based on phase amplification.
 - 3.5 Algorithms based on the quantum Fourier transform.
4. **Part 3: Quantum programming in PennyLane.**

Our home

<https://lmf.di.uminho.pt/FoundationsofQC2526/>